

10. Electroencephalogram (EEG) is used in monitoring :

- (a) Heart (b) Liver
(c) Pancreas (d) Brain
(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (d)

An EEG is a test that detects electrical activity of the brain using small, metal discs (electrodes) attached to the scalp. Brain cells communicate via electrical impulses and are active all the time, even when a person is sleeping. This activity shows up wavy lines on an EEG recording.

11. For a satellite, escape velocity is 11 km/sec. If the satellite is launched at an angle of 60° with the vertical, then the escape velocity will be

- (a) 11 km/sec
(b) $11\sqrt{3}$ km/sec
(c) $11/\sqrt{3}$ km/sec
(d) 33 km/sec
(e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

Escape velocity is not an actual velocity at all. It's speed is a scalar quantity, so its value remains unchanged at any angle.

MECHANICS

Mass, Motion, Force, Acceleration, Momentum, Work, Power, Energy

1. What is the law in which under the same conditions of temperature and pressure, equal volumes of all gases contain equal number of molecules?

- (a) Coriolis effect (b) Graham's law
(c) Pascal's law (d) Avogadro's law
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

At the same temperature and pressure all the ideal gases have the same number of particles or molecules in the same volume. This is the Avogadro's law.

2. A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to

- (a) decrease the rate of change of momentum
(b) exert large force on the ball

- (c) increase the force exerted by the ball on hands
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

When a goalkeeper pulls his hands backward after holding the ball shot at the goal, he is effectively increasing the time over which the change in momentum occurs.

This action helps reduce the rate of change of momentum and, consequently, the force experienced by the goalkeeper's hands.

The principle involved in this is known as impulse-momentum theorem, which states that the change in momentum of an object is equal to the impulse applied to it.

3. When a stone is brought from the surface of the moon to the earth, then :

- (a) Its mass will change
(b) Its weight will change, but mass will not change
(c) Both weight and mass will change
(d) Neither mass nor weight will change

46th B.P.S.C. (Pre) 2004

Ans. (b)

When a stone is brought to the earth from the surface of the moon, its mass remains constant but the weight will increase because the acceleration due to gravity is Higher on the earth than on the moon.

4. Which of the following is not a vector quantity?

- (a) Displacement (b) Velocity
(c) Force (d) Volume

45th B.P.S.C. (Pre) 2001

Ans. (d)

The scalar quantity is defined as the quantity that has only magnitude, whereas, vector quantity comprises of both magnitude and directions. e.g. Displacement, Velocity, Torque and Force are the examples of vector quantity while volume, speed are scalar quantity.

5. Which of the following is not a vector quantity?

- (a) Speed (b) Velocity
(c) Torque (d) Displacement
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

See the explanation of the above question.

6. Which one of the following is a scalar quantity?

- (a) Force (b) Pressure
(c) Velocity (d) Acceleration
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Force, velocity and acceleration are the examples of vector quantity while pressure is a scalar quantity. Pressure is defined as the force per unit area. It is the force acting on a unit area of surface while force is a pull or push in a specific direction.

7. Current density is

- (a) a scalar quantity (b) a vector quantity
(c) dimensionless (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

Current density is a vector quantity because it is a product of charge density and velocity.

8. A sudden rise in the barometric height indicates

- (a) coming of storm or cyclone
(b) possibility of rain
(c) dry weather
(d) extremely dry weather
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (d)

A barometer measures air pressure: A "rising" barometer indicates increasing air pressure; a "falling" barometer indicates decreasing air pressure. When the air is dry, cool, and pleasant, the barometer reading rises. In general, a rising barometer means improving weather. A rapid rise indicates unsettled weather, with extremely dry weather conditions.

9. The time period of a pendulum clock is

- (a) 1 second (b) 2 seconds
(c) 1 minute (d) 1 hour
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

Time period is the time taken to cover one oscillation. When a pendulum clock goes from one position to another it takes 1 sec.

10. Which physical quantity is represented by the ratio of momentum and velocity of the body?

- (a) Velocity (b) Acceleration

(c) Mass

(d) Force

45th B.P.S.C. (Pre) 2001

Ans. (c)

Momentum is a vector quantity that is the product of mass and the velocity of an object or particle.

Its direction is the same as that of velocity. The SI Unit of momentum is kilogram-meter per second (kg m s^{-1})

Momentum = Mass $\times$ Velocity

$$\text{So, Mass} = \frac{\text{Momentum}}{\text{Velocity}}$$

11. Which is the correct formula to find the acceleration?

- (a) $a = (v-u)/t$ (b) $a = u + vt$
(c) $a = (v + u) / t$ (d) $a = (v + u) / 2$

45th B.P.S.C. (Pre) 2002

Ans. (a)

$$\text{Acceleration} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time Difference}}$$

$$a = \frac{v - u}{t}$$

12. Force is the product of –

- (a) Mass and velocity (b) Mass and acceleration
(c) Weight and velocity (d) Weight and acceleration

45th B.P.S.C. (Pre) 2001

Ans. (b)

A force can cause an object with mass to change its velocity (which includes to start moving from a state of rest and vice versa) i.e., to accelerate.

Force (F) = Mass $\times$ Acceleration

13. Ball bearings are used to convert static friction into

- (a) rolling friction (b) drag
(c) sliding friction (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Ball Bearings are used to convert static friction into rolling friction. Static friction occurs, when two surfaces are in contact but not sliding past each other.

The rolling motion of the ball bearings enables smoother movement between the two surfaces, reducing the frictional resistance significantly.

14. Which of the following energy changes involves frictional force?

- (a) Kinetic energy to heat energy
(b) Potential energy to sound energy